



SEMESTER 2 2025/2026

FACULTY: FTKEK

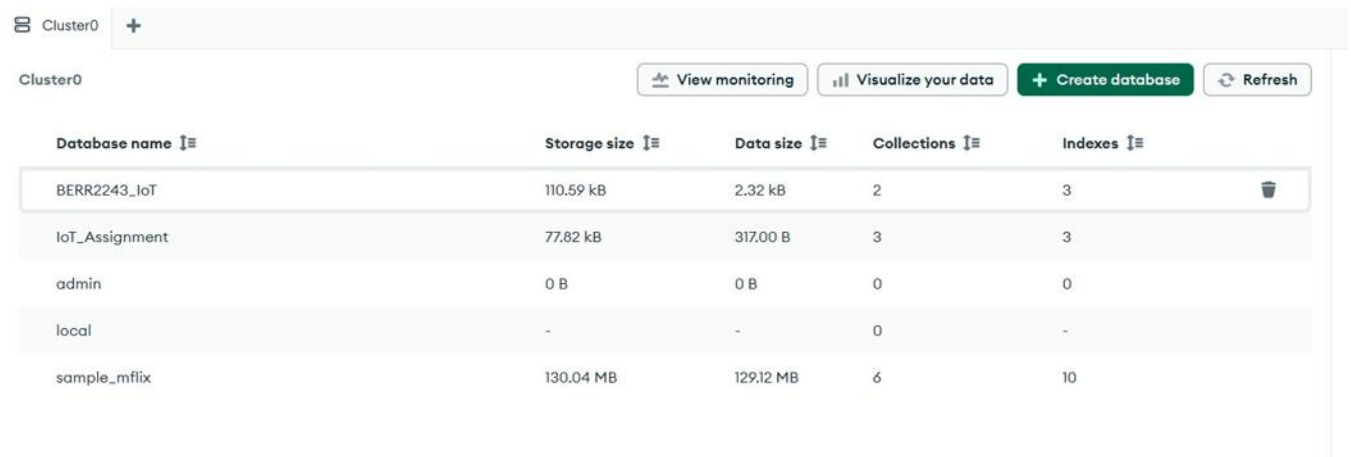
BERR2243 : DATABASE AND CLOUD SYSTEMS

TITLE: End-to-End Cloud IoT Platform Deployment

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DEPLOYMENT PROCESS



The image shows the MongoDB Atlas cluster overview for Cluster0. At the top, there are buttons for 'View monitoring', 'Visualize your data', 'Create database', and 'Refresh'. Below these is a table listing databases and their metadata.

Database name	Storage size	Data size	Collections	Indexes
BERR2243_IoT	110.59 kB	2.32 kB	2	3
IoT_Assignment	77.82 kB	317.00 B	3	3
admin	0 B	0 B	0	0
local	-	-	0	-
sample_mflix	130.04 MB	129.12 MB	6	10

Figure 1

Figure 1 shows MongoDB Atlas cluster overview showing the BERR2243_IoT database with the metadata and telemetry collections successfully created for storing device authorization and time-series sensor data.

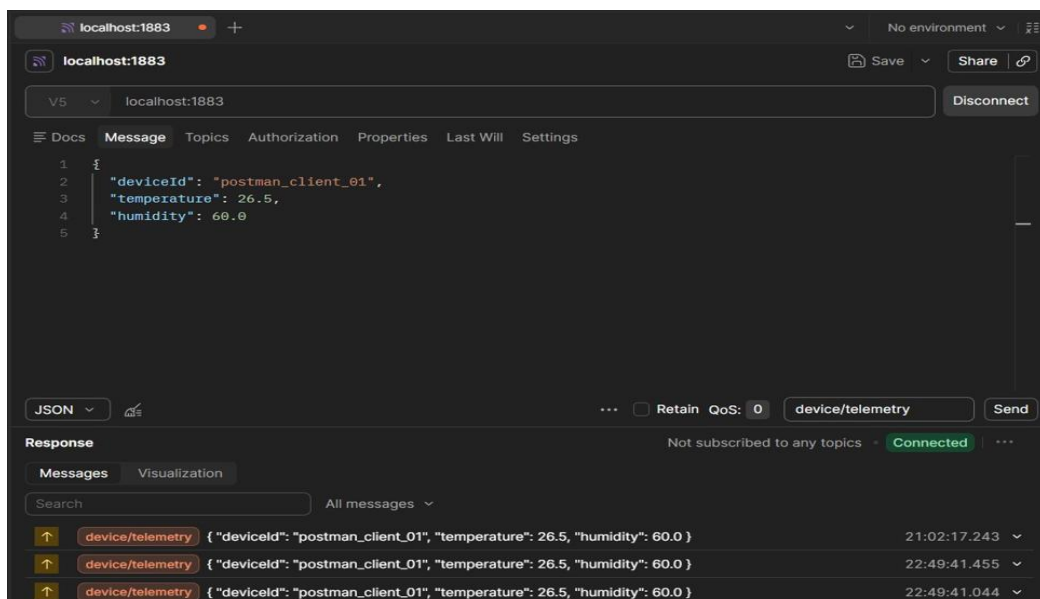


Figure 2

Figure 2 shows the Time-series telemetry data stored in MongoDB Atlas, confirming the successful ingestion of JSON payloads (deviceId, temperature, humidity, timestamp) from the MQTT broker

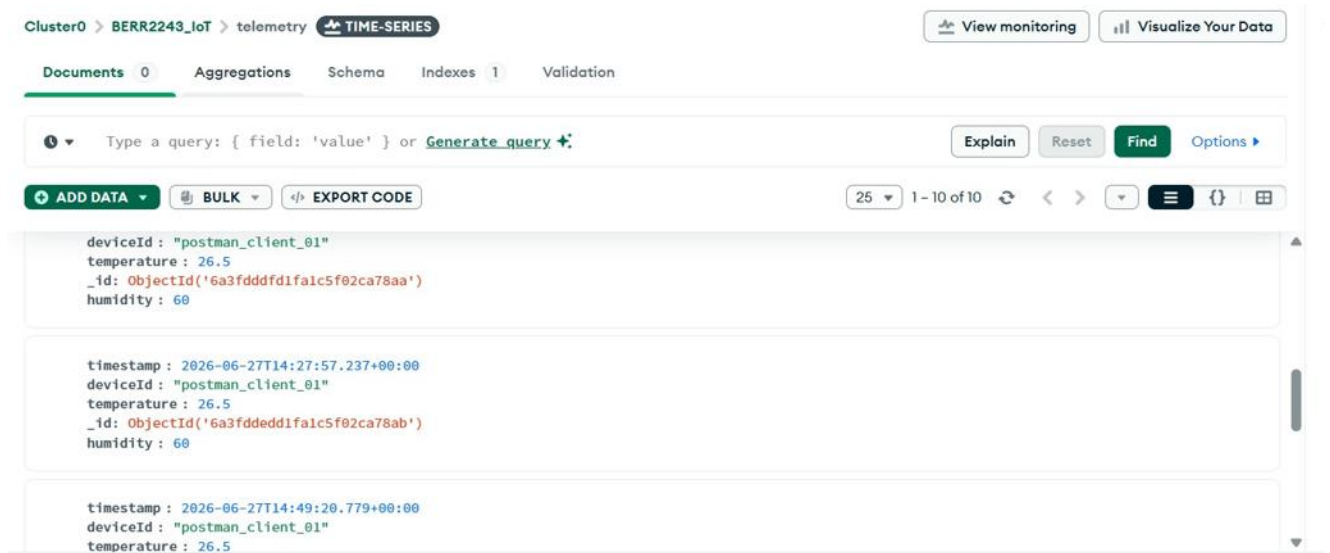


Figure 3

Figure 3 shows the MQTT client publishing telemetry data to the broker at localhost:1883 on topic device/telemetry. The JSON payload matches the format required by the database, with a QoS level of 0. The message history confirms successful transmission.

SECURITY PROTOCOL



Figure 4

Figure 4 shows the device authorization metadata collection. The system checks this collection to verify if a device is authorized (authorized: true) before allowing telemetry data to be written to the time-series collection, demonstrating database-level security validation.

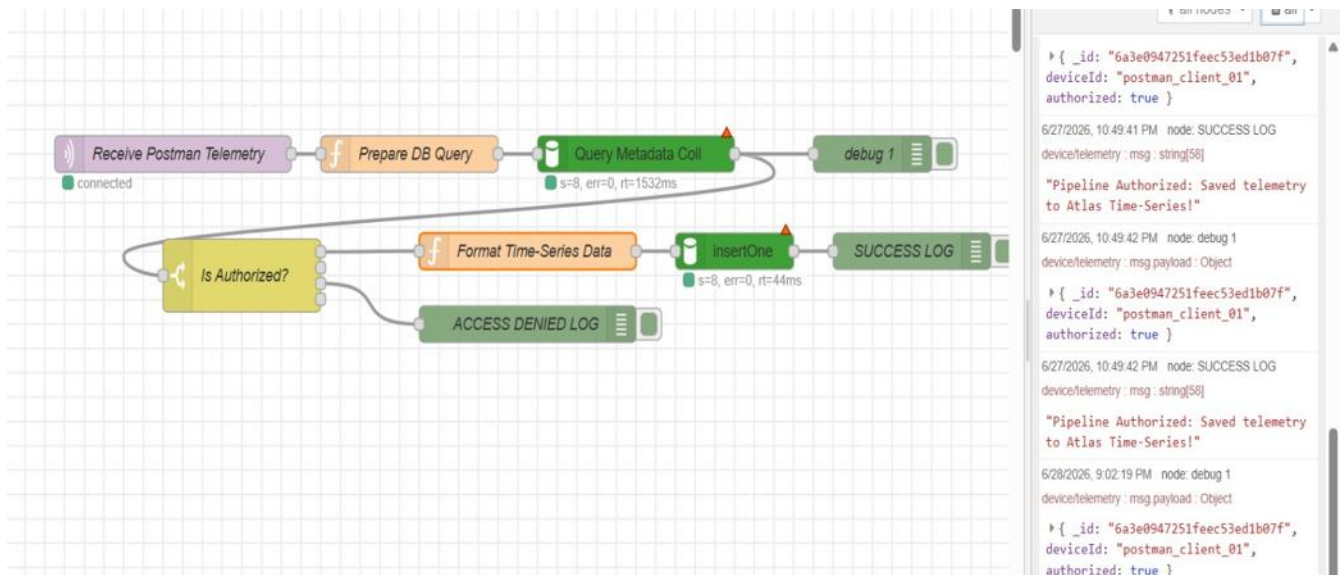


Figure 5

Figure 5 shows the Node-RED flow execution logs demonstrating the authorization pipeline. The flow queries the metadata collection for device authorization status, and upon successful validation (authorized: true), formats and saves the time-series data to MongoDB Atlas. The debug nodes confirm payload processing at each stage.

CONCLUSION

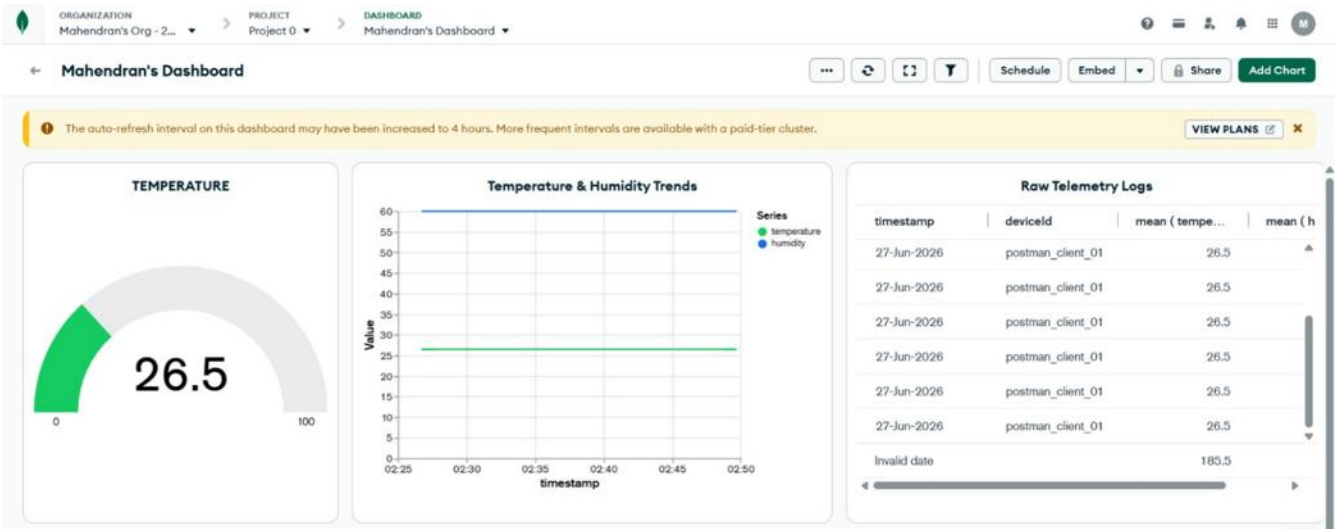


Figure 6

Figure 6 shows the Final IoT dashboard displaying real-time temperature (26.5°C) and humidity (60%) trends, along with raw telemetry logs. The auto-refresh interval is configured for data visualization, completing the end-to-end pipeline from ESP32 → MQTT Broker → Node-RED → MongoDB Atlas → Dashboard.